Manish Tiwari

AI Engineer | LLM Systems, Agent Tooling & Automation | Open Source

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Summary

AI engineer building the infrastructure and services LLM applications run on: sandboxed execution, agent tooling, guardrails and observability. Ships production Python end to end with FastAPI, Docker and CI/CD. 41 merged pull requests across open-source organisations; 3 packages on PyPI.

Experience

DALM Labs Remote

AI Engineer (Contract) *Oct 2026 -Present*

- Built the AI planner behind a CPaaS audience studio -a FastAPI service turning a domain, objective and prompt into an audience recommendation, signal selection with per-signal reasoning, and a budget plan, with Sarvam and Claude behind a provider-agnostic interface.
- Integrated it with the platform's tenant and trait services: tenant resolved from the caller's JWT and forwarded downstream, warehouse traits mapped into the planner's signal vocabulary with per-trait coverage and reach, cached and retried, with a static fallback when the platform is unavailable.
- Enforced LLM guardrails - signals outside the resolved vocabulary dropped server-side, budget channels restricted and renormalised to **100%**, schema validation with one corrective retry - and shipped a **58-test** suite; fixed three production bugs that surfaced only against a live model.

Smartivity Labs Pvt. Ltd. New Delhi

Automation Intern *June 2026 - August 2026*

- Built an end-to-end Slack-integrated project platform used by **100+ designers and managers** across **200+ projects** - Slack OAuth (bot install + OIDC login with PKCE), automated per-project channels, timezone-aware reminders, and an in-channel approval workflow that auto-advances project state, with analytics on overdue projects and completion trends. Replaced a 100-user spreadsheet and saves **hundreds of minutes each week**.
- Automated certificate generation and PPT/PDF → Excel extraction pipelines, cutting processing time by **over 90%** - ∼1 minute to **10–12 seconds** per certificate, and a 2-3 hour manual workflow down to a few minutes.

Open Source Software Contributor Global

Hugging Face, Augur, Moss *Jan 2024 -Present*

- Contributed 41+ merged PRs across 10+ open-source organizations including Hugging Face, Moss, and CHAOSS/Augur, shipping SDK improvements, automated tests, developer tooling, documentation, and AI application features.
- Built Bharat Benefits, a voice-first RAG assistant merged into Moss, integrating Sarvam AI speech models and Moss retrieval implementing an end-to-end STT → Retrieval → LLM → TTS pipeline for Indian government welfare scheme discovery.

Travintop.com Remote

Data Operations Intern *Dec 2024 -Feb 2025*

- Built Python automation scripts to process and normalize 500+ travel package records for CMS upload workflows.
- Analyzed influencer engagement metrics and structured operational datasets supporting marketing and listing operations.

Projects

mesh, Distributed GPU Training Planner [GitHub] [PyPI] 2026

- Built a capacity planner that computes the real training ceiling per node across a heterogeneous GPU fleet, replacing summed pooled VRAM: on 100 consumer GPUs, only **8** can hold a 7B model, a limit that a month of failed runs used to reveal.
- Implemented the DiLoCo outer loop with int8 pseudo-gradients and churn-tolerant barriers, auto-tuning sync interval to keep upload under **20%** of compute; shipped CLI, coordinator and a live fleet dashboard with measured throughput and ETA.

SafeRun AI, Secure Sandbox Execution Platform [Package] [GitHub] [Live] May 2026

- Built a Docker sandbox executing untrusted AI-generated Python with CPU, memory and timeout limits; AST scanner and YAML policy engine classify execution into **4 risk levels**.
- Published as **saferun-ai** on PyPI and shipped as an **MCP server** exposing scan, execute and history tools to any MCP client; audit logging, LLM explanations and browser UI.

LocalMind, Local LLM Hardware Advisor [PyPI] [GitHub] 2026

- Shipped an **18-command** CLI, published as **localmind-ai** on PyPI, that scans CPU, RAM and GPU and recommends which local LLMs will actually run, with VRAM estimates at every quantization from FP16 to Q2_K.
- Implemented multi-vendor GPU detection (NVIDIA, AMD, Intel, Apple Silicon) with tier classification treating Apple unified memory differently from Windows RAM; added Ollama search, auto-install and chat.

Laya on a GTX 1650, LLM Inference Benchmark [GitHub] [Blog] Sep 2026

- Benchmarked a **421M**-parameter LLM on 4GB consumer hardware with CUDA-synchronised timing, 5 warmups and **100** timed calls per scenario, reporting p50/p95 and separating measured results from third-party published figures.
- Found latency **15 to 22x** slower than published T4 numbers, a **3.8x** per-question cost drop from batching, and a 50-question batch that exceeded VRAM (mean 2.4x median), flagging batch-size risk for 4GB deployments.

Technical Skills

Automation: Python, Workflow Automation, Slack Integrations, Document Processing, Playwright
AI/ML: PyTorch, HuggingFace, Transformers, LangChain, OpenAI SDK, RAG, LLM Workflows, LoRA, Fine-Tuning
Backend/Infra: FastAPI, REST APIs, Docker, Linux, GitHub Actions
Tools: Streamlit, Git&Github, Ollama, Plotly, SentencePiece, Scikit-learn

Education

IK Gujral Punjab Technical University Jalandhar, Punjab, India
Bachelor of Technology in Computer Science & Engineering; CGPA: 8.0 *July 2023 – June 2027*

Achievements

- Trained **Trimurti-LM**, a 4.2M-parameter multilingual GPT (EN/HI/PA), from scratch to perplexity **3.32** in **2.38 hours** on a 4GB GTX 1650 using FP16, gradient checkpointing and gradient accumulation.
- Built and published Ollama libraries for Sarvam 30B and 105B multilingual LLMs, **100+ community pulls**.
- Won Industrial Automation hackathon category for building **EnergiX Copilot**, an AI assistant for energy optimization and operational monitoring; also fine-tuned Mistral-7B & Qwen-14B using LoRA for dual-agent workflows at AMD Hackathon, IIT Delhi.